SOMAIYA VIDYAVIHAR UNIVERSITY

16-05-2025 (E)

Semester: January 2025 –May 2025		
Maximum Marks: 100 **Examination: ESE Examination**		**Duration: 3 Hrs.**
Programme code: 42 **Programme: Minors in Augmented and Virtual reality**	**Class: TY**	**Semester: VI (SVU 2020)**
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department: All** COMP/ETRX/EXTC/IT/MECH/RAI/AIDS/CCE/EXCP____	
Course Code: 116m42C601	**Name of the Course: Augmented Reality and Multidimensional System (4DX, DTS)**	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **four**	**20**
i)	Application of Multidimensional Systems in Industry.	5
ii)	Write a short note on the introduction to AR and 3D systems.	5
iii)	Explain the working of Cave systems.	5
iv)	Explain Sound mixing and Multichannel video synchronization.	5
v)	Write simple code in C# programming to display Hello World.	5
vi)	Explain the Antialiasing setting.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Explain Motion graphics software	5
ii)	How AR is used in Healthcare	5
	OR	
Q2 A	Explain the working of the 4D-X, Transformers ride in Universal Studios	10
Q 2 B	Solve any **one**	**10**
i)	Compare Google glass with Magic leap one	10
ii)	Compare Pneumatic controllers with Hydraulic controllers	10

Que. No.	Question	Max. Marks
Q3	Solve any **two**	**20**
i)	Write a short note on Augmented reality softwares.	10
ii)	Explain the Luminance Channel with one example.	10
iii)	Explain the virtual photo studio.	10

Que. No.	Question	Max. Marks
Q4	Solve any **two**	**20**
i)	How AR is used for Gaming, explain with a Case study of Pokémon GO	10
ii)	Explain 4D-X movie-making software	10
iii)	Explain the working of Electric servos	10

Que. No.	Question	Max. Marks
Q5	Write a short note on any **four**	**20**
i)	Flight simulators	5
ii)	Bus/Truck simulators	5
iii)	Mining simulator	5
iv)	Parachute Simulator	5
v)	GearAR	5
vi)	The Fog channel	5